AV AI VELOCITY ARCHITECT INTELLIGENCE · ISSUE 01

WEEK OF 29 AUGUST – 5 SEPTEMBER 2026

Where The Signature Lives

The human step in the agent loop moved four times in one week: into the tool protocol, into the model turn, onto a call identity and under a lock file.

2	2	7	12
repos run	mechanisms executed	papers read	models priced

Vijay Zharotia
Healthcare AI architecture

WHY AI VELOCITY

Agents that act inherit obligations.

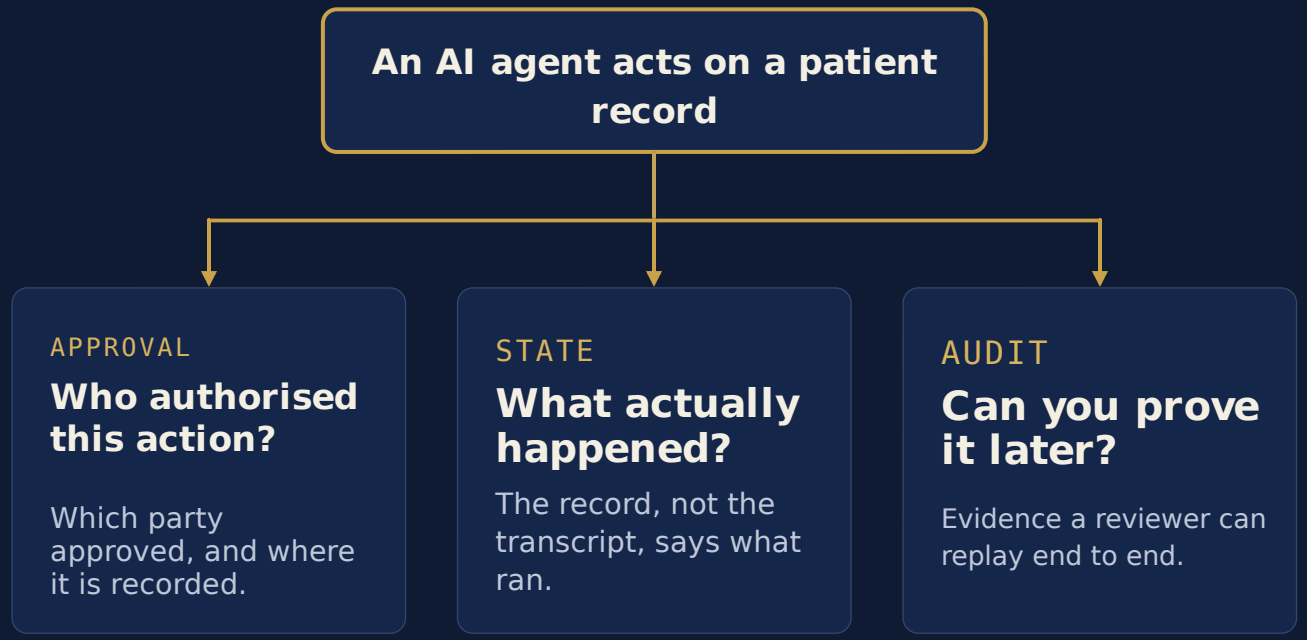
A weekly architecture brief for teams putting AI agents into healthcare workflows. We read the releases, run the mechanism, and say what to change.

FOR ARCHITECTS

The mechanism, the code, the move.

FOR PRODUCT MANAGERS

The decision, the risk, the date.



EVERY CLAIM CARRIES ITS EVIDENCE

EXECUTED

We ran it. The numbers are observed output.

DOCS

Read from documentation or source.

UNVERIFIED

Not confirmed. Vendor numbers carry an asterisk.

THE WEEK IN ONE VIEW

The human step moved four times in one week

CONTEXT

The signature lived in your client

An agent proposed an action, your code asked a person, and the approval sat between the model and the tool.

→

PROBLEM

Four frameworks moved it

Into the tool protocol, into the model turn, onto a stable call identity and under a lock file. No release note said so.

→

SOLUTION

Pick one home for the signature

Every new place a signature can live is a new place the audit transcript must be written. Choose one; make the others defer to it.

SIGNATURE · 04–06
Where a human can sign

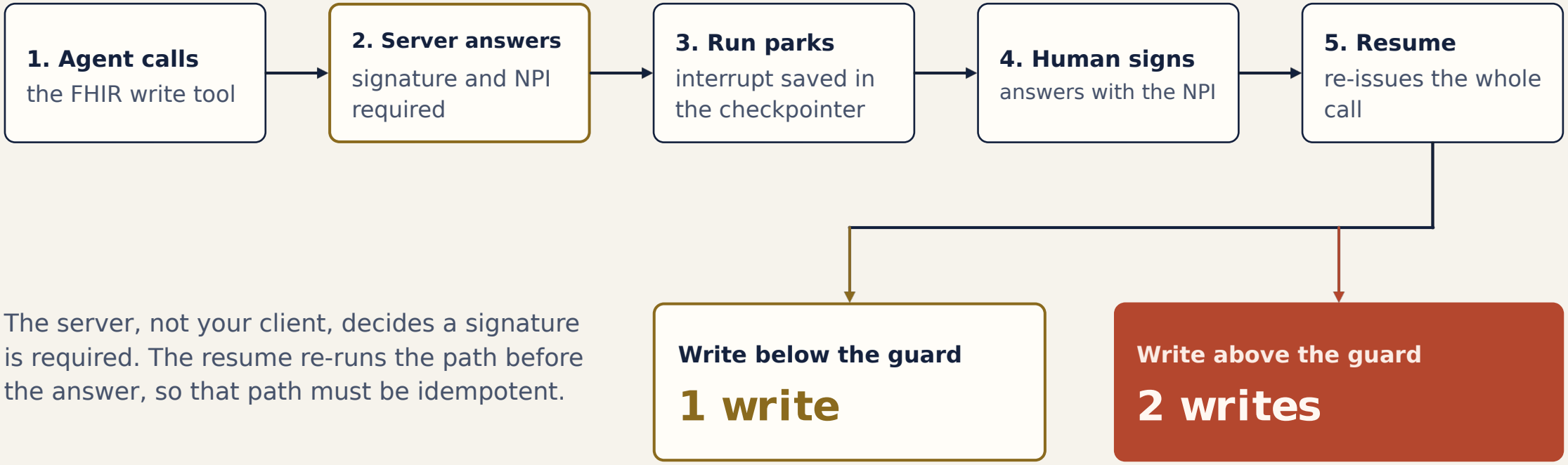
CONTROL · 07
Proving the validated agent

COST · 08
10,000 documents

EVIDENCE · 09
Results to act on

EXECUTED MECHANISM · PROBLEM · LANGCHAIN 1.4.0 + FASTMCP

The signature that lives in the tool protocol



The server, not your client, decides a signature is required. The resume re-runs the path before the answer, so that path must be idempotent.

EXECUTED MECHANISM · SOLUTION · YOUR HARNESS

Let the server own the gate; make the resume safe

STEP 1

Server demands the signature

Every elicitation-capable client gets the same gate.

→

STEP 2

Persist the run

State is mandatory; the interrupt lives in the checkpointer.

→

STEP 3

Make the pre-answer path idempotent

The resume runs it again.

→

STEP 4

Write once, below the guard

Only after the signature lands.

0

writes before the signature

1

write after the resume

1

advertised flag that is not in the code

> The pull request advertises an elicitation flag on the adapter; the merged code and the wheel have none. Interrupt-driven elicitation is on by default.

Two new places a human can sign

CONTEXT
how it runs

Model proposes a write → **Approval bound to a call** → **Model turn continues** → **Tool result arrives** → **Write to FHIR**

PROBLEM
where it broke

MS AGENT FW 1.17
A forged id failed closed; the model still said done
EXECUTED

OPENAI ASYNC
A turn can end before your tool does
DOCS

OPENAI STEERING
A steered turn stops; a successor continues
DOCS

SOLUTION
what to build

Write the audit transcript in one place

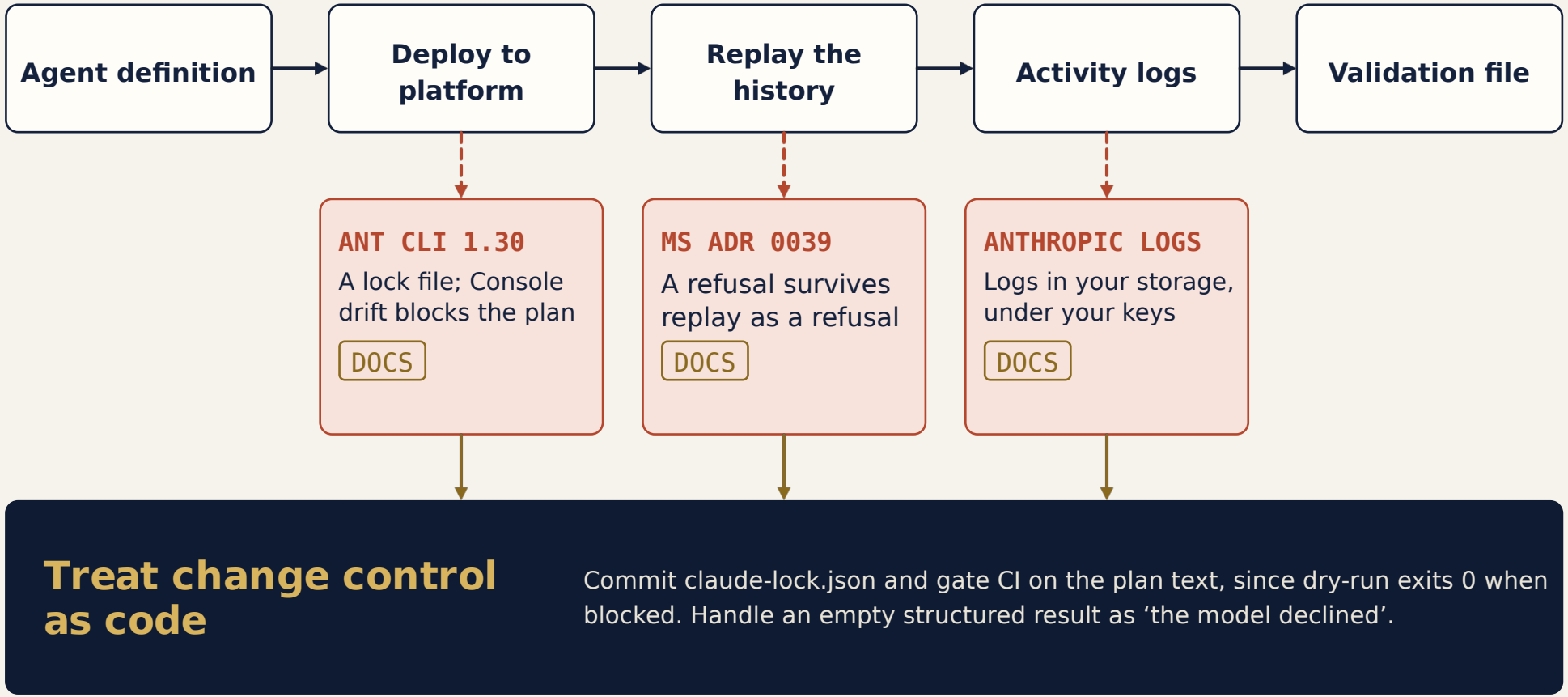
Store the occurrence id and the provider id, keep an idempotency key at the FHIR layer, track in-flight call ids, and never read success from the model’s text.

Proving the agent is the one you validated

CONTEXT
how it runs

PROBLEM
what moved

SOLUTION
what to build



COST · LIST PRICES ON 5 SEPTEMBER

The same extraction job for $48.75, not $650

MODEL	COST PER 10,000 DOCS	INDEX*
Gemini 3.8 Flash	**$48.75; $97.50 from 1 Jan 2027**	**59**
Claude Opus 5	**$325**	not given
Claude Fable 5.1	**$650**	**66**
GPT-6 Astra	**$650**	**61**

Per million tokens in/out: Flash $0.75/$3.75 through 31 Dec, then $1.50/$7.50; Opus 5 $5/$25; Fable 5.1 and Astra $10/$50. 4,000 in and 500 out per document, no caching or retries.
*Artificial Analysis Intelligence Index v4.2 as recorded 5 Sep, pre-rebase, not re-checked; not comparable with later issues.

CONTEXT

Extraction work is often bound to one flagship model.

PROBLEM

For this job shape that costs 13 times a bounded-extraction model.

DECISION

Budget extraction on Flash with the January doubling written in; keep Opus 5 for reasoning. On-premise PHI: the licence decides, so pilot Apache 2.0 Granite 4.2 8B or Qwen3.8-27B.

Three results, one boundary to draw

CONTEXT
how it runs

Visit history → **Agent memory layer** → **LLM judge on notes** → **Human review** → **Governance sign-off**

PROBLEM
where it broke

2609.01111
0–5.3%
of cross-visit relations kept by memory layers

2608.31016
0.50–0.63
omission detection; additions 0.79–0.94

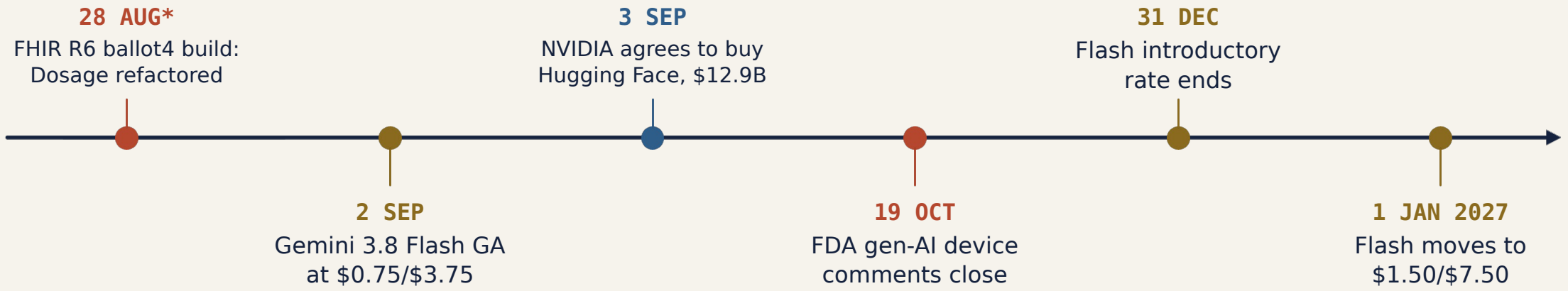
2609.02095
39 vs 30%
review for agents 0.3 points apart

SOLUTION
what to build

Verify outside the model

Use full context or a verified timeline, enumerate transcript facts and check each one, and report oversight burden at target reliability, not accuracy.

Dates that decide what you build on next



● REGULATION, STANDARDS ● PRICE ● SUPPLY

*As reported in the issue; not confirmed on hl7.org.

What reaches your backlog before any model

LLAMAINDEX

A long-document cliff

Gemini 3.5 Flash drops from 87.9% on short documents to 27.9% on long ones in ExtractBench.

PYDANTIC

Realtime speech agents

Typed server-side tools; 2.40.0 adds barge-in and out-of-band prompts for escalation mid-call.

SIMON WILLISON

An egress proxy bypassed

Research agents edited /etc/hosts. Network policy for a PHI agent belongs outside the container.

HUGGING FACE

Owned local agent memory

Funes on Lance; a GRPO recipe lifts a 350M model from 18% to 32% valid JSON.

OPENAI HEALTH

ChatGPT connected to Epic

Read-only at launch, per OpenAI. The transport, SMART on FHIR or proprietary, is not disclosed.

HL7

FHIR R6 ballot4 build*

Dosage refactored across MedicationRequest, Dispense and Statement.

Seven moves, with the date that forces each

Move	Why	By
Store occurrence and provider ids; idempotency key at FHIR	Replays can re-arm approvals	Next release
Track in-flight async call ids and the latest response id	A turn can end before the tool	Before async
Commit claude-lock.json; gate CI on the plan text	Dry-run exits 0 when blocked	This sprint
Budget extraction on Flash with the January doubling	$48.75 against $650	Before 1 Jan 2027
Mirror the open weights you depend on	Hugging Face changing owner	Now
Replace holistic scribe judges with fact enumeration	Omissions caught at 0.50–0.63	Next QA cycle
File an FDA comment if you ship SaMD on a foundation model	Docket FDA-2026-N-7874	19 Oct

AV AI VELOCITY

THIS WEEK'S POSITION

Pick one place the signature lives. Make the others defer to it.

Long-form edition
Every mechanism, source and date

Lab package
Runs clean, no API key needed

New issue every Friday
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github.com/zkumar/ai-velocity-weekly